Challenge 1 - Psychophysics Prototype

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Setup addressed

- #1, Visual vs. Auditory Reaction Time

Research question

Do people react faster to **light** or to **sound**?

A participant presses a button as soon as an LED lights up (visual) or a buzzer sounds (auditory). Both stimuli are mixed in one session, so we can compare them under the same conditions.

Tools used

- Arduino + Grove base shield
- Grove push button, LED, Grove buzzer, wires
- Arduino IDE (used only to upload StandardFirmata)
- StandardFirmata (runs on the Arduino)
- Processing with its built-in Serial library (runs the experiment on the laptop)

Wiring and Arduino

Wiring

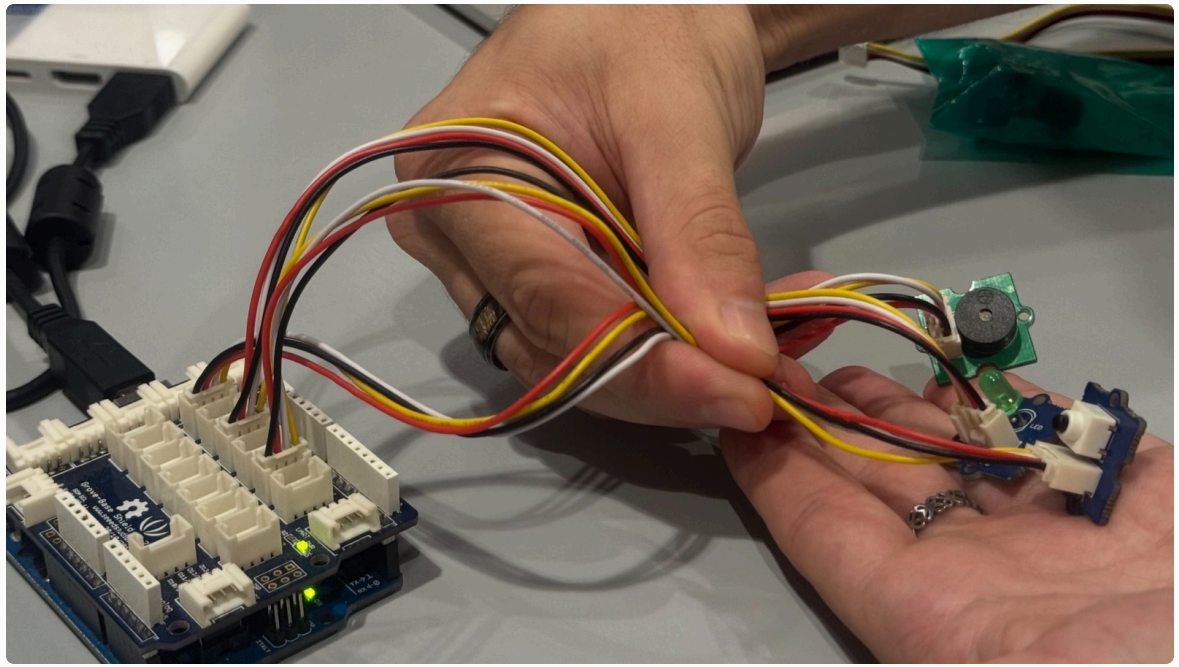
- **Button:** *Digital port 2 (D2)*, a Grove button module. The pin uses the internal pull-up, and the module outputs HIGH when pressed.
- **LED:** *Digital port 4 (D4)* → 220 ohm → LED anode, cathode → GND
- **Buzzer:** *Digital port 6 (D6)* → buzzer (+), buzzer (-) → GND. This is an active buzzer: it beeps while the pin is HIGH.

On the Arduino

- We uploaded StandardFirmata once. The board runs no experiment code. It only switches pins when told to and reports pin changes to the laptop (57600 baud).
- All the logic (trial order, random wait, timing, saving data) runs in Processing.

Pin modes

Button, pin 2: input with pull-up
LED, pin 4: digital output (on / off)
Buzzer, pin 6: digital output (on / off)



Switching to Processing

Talking to the Arduino

- The Firmata library did not connect for us, so we send the Firmata commands ourselves with Serial.write.
- At start-up: SET_PIN_MODE (`0xF4`) for each pin, then REPORT_DIGITAL (`0xD0`) so the board reports button changes.
- LED and buzzer: switched on and off with the digital-pin command SET_DIGITAL_PIN (`0xF5`).
- Button: serialEvent() saves System.nanoTime() the moment a press arrives (pin 2 goes from LOW to HIGH).
- We decided to use nanoTime() rather than millis() because it gives sub-millisecond values instead of rounded timestamps, and the press time doesn't depend on the frame rate.

Running the experiment

- 10 visual and 10 auditory trials, in random order. **L** / **B** test the LED and buzzer, and **SPACE** starts.
- 800 ms pause, then a random wait of 1–4 s, then the LED or buzzer turns on. Reaction time = press time − stimulus time.
- If press before the stimulus = false start. If no press within 2 s = timeout (trial is invalid).
- Each trial is saved as a row in a CSV (stimulus, wait, reaction time, outcome). At the end, the median and mean for each stimulus are shown on screen.